

- Consider the Classical solⁿ of a string w/ length L , the Energy E and angular Momentum $J \equiv |\vec{J}^{12}|$ are given by

$$E = \frac{2}{4\alpha'} L$$

(Energy)

$$J = L^2 / 16\alpha'$$

(Angular Mom.)

→ Rotating string

$$L \Rightarrow \alpha' E^2 = J$$

* When we Quantize the theory we have

$$J = n\hbar \quad \text{and so}$$

$$\alpha' E^2 = n\hbar, \quad E = [\text{Rest mass}]$$

⇒ We have a series of masses m_n

$$\alpha' m_n^2 = n\hbar$$

We have an ∞ # of particles whose m_n^2 is proportional to their spin J .

The Old Covariant Method:

$$\left[\begin{array}{l} \text{Poisson} \\ \text{Dirac} \end{array} \right] \quad \left\{ \begin{array}{l} \{A, B\} = C \\ [A, B] = i\hbar C \end{array} \right.$$

→ Can always be applied to phase space coordinates

(i) Momenta: $P^\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

In the (conformally) flat gauge $g^{\alpha\beta} = \eta^{\alpha\beta}$

$$P^\mu = \frac{1}{2\alpha'} \dot{x}^\mu$$

• We found $\{x^\mu(\sigma), \dot{x}^\nu(\sigma')\} = \sigma \{P^\mu(\sigma), P^\nu(\sigma')\}$

$$\{x^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

Which are all classical constraints; thus, we can impose the Quantum Dirac relations:

$$[x^\mu(\sigma), P^\nu(\sigma')] = i\hbar \eta^{\mu\nu} \delta(\sigma - \sigma')$$

$$[X^M(\sigma), X^N(\sigma')] = 0 = [P^M(\sigma), P^N(\sigma')] = 0$$

$$[X^M(\sigma), P^N(\sigma')] = i\hbar \delta(\sigma - \sigma') \eta^{MN}$$

* The Schrödinger representation:

$$X^M(\sigma) \mapsto \hat{X}^M(\sigma)$$

$$P^M(\sigma) \mapsto -i\hbar \frac{\delta}{\delta X^M(\sigma)}$$

where

$$\delta X^N(\sigma) / \delta X^M(\sigma') = \delta^N_M \delta(\sigma - \sigma')$$

Constraints:

(i) Open string: $T_{\alpha\beta} = 0$ or $L_n = 0$
why?

$$T(z) = \sum_{n \in \mathbb{Z}} L_n z^{n-2}$$

and since $T(z) = 0$ and $T(\bar{z}) = 0$

then $T(z) = 0 \Leftrightarrow L_n = 0$

(ii) Closed string: $L_n = 0$ and $\bar{L}_n = 0$

These are the constraints which we impose on a wave f.e. $\psi[X^M(\tau, \sigma)] \equiv \langle X^M(\sigma, \tau) | \psi \rangle$

The Open string:

* In extending $\sigma \rightarrow [-\pi, \pi]$ we combined both constraints into a single condition

$$L_n = 0 \quad \forall n$$

(Recall $X_{-n}^M = X_n^M$, and so " $\bar{L}$ " = L here).

- A constant must be added to L_0 to parametrize the arbitrariness of the ordering chosen. So, at best we may have $(L_0 - a)\psi = 0$

or, normalizing $a=1$

$$(L_0 - 1)\psi = 0, \quad L_n \psi = 0, \quad n \geq 1$$

(Physical state conditions)

$$L_n^\dagger = L_{-n}, \text{ it's clear } \langle \psi | L_n | \psi \rangle = 0, \quad n \neq 0$$

$$\bullet \quad \dot{x}^\mu, p^\mu \leftrightarrow q^\mu, \alpha_0^\mu = \sqrt{2\alpha'} p^\mu, \alpha_n^\mu, \quad n \neq 0$$

$$\bullet \quad \text{Use } \tau=1, \quad \{\alpha_n^\mu, \alpha_m^\nu\} = -in \delta_{n+m,0} \eta^{\mu\nu}$$

and so $\{ \} \rightarrow [\cdot, \cdot] \propto -i$

$$[\alpha_n^\mu, \alpha_m^\nu] = n \delta_{n+m,0} \eta^{\mu\nu}$$

36, 52

$$x_0^\mu(\tau) = \frac{1}{\pi} \int_0^\pi d\sigma^\mu x^\mu(\tau, \sigma)$$

$$\{x^\mu(\sigma), p^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

Integrating over σ : at $\tau=0$

$$\left\{ \frac{1}{\pi} q^\mu, p^\nu \right\} = \eta^{\mu\nu}$$

$$\int_{-\pi}^{\pi} d\sigma^\mu \{ \cdot, \cdot \} = \eta^{\mu\nu}$$

$$\{ q^\mu, p^\mu \} = \eta^{\mu\nu}$$

$$\Rightarrow \{ q^\mu, p^\mu \} = \eta^{\mu\nu}$$

$$\text{Thus: } [q^\mu, p^\mu] = i\eta^{\mu\nu}$$

This is embedded in the ladder ops

$$\alpha_0^\mu = \sqrt{2\alpha'} p^\mu = -i\sqrt{2\alpha'} \frac{\partial}{\partial q^\mu}$$

$$x_0^\mu = q^\mu$$

Since $p^\mu(\sigma)$ is self adjoint $p^{\mu\dagger} = p^\mu$,

$$\text{then } (\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$$

* We can redefine these relations as follows:

$$\begin{aligned} \dot{x}^\mu(\sigma) &= \sqrt{2\alpha'} p^\mu + \frac{x^{\mu'} / \sqrt{2\alpha'}}{\sqrt{2\alpha'}} \\ &= -i\sqrt{2\alpha'} \frac{\partial}{\partial x_{\mu}(\sigma)} + \frac{x^{\mu'}(\sigma)}{\sqrt{2\alpha'}} = \sum_{n=-\infty}^{\infty} \alpha_n^\mu e^{in\sigma} \end{aligned}$$

$$\delta \sum_n \frac{\partial x_n^\mu}{\partial} \frac{\partial}{\partial}$$

$$\delta X^\mu(\sigma) = \sum_{n=0}^{\infty} \partial X^\mu(\sigma) \partial X_n^\mu$$

Since $X_n^\mu(\tau) = \frac{2}{\pi} \int_0^\pi d\sigma' \cos(n\sigma') X^\mu(\tau, \sigma')$

$$X_0^\mu(\tau) = \frac{2}{\pi} \int_0^\pi d\sigma' X^\mu(\tau, \sigma')$$

$$\sum_{n=0}^{\infty} \frac{2}{\pi} \int_0^\pi d\sigma' \cos(n\sigma') \delta X_n^\mu \delta(\sigma - \sigma')$$

$$= \sum_{n=0}^{\infty} \frac{2}{\pi} \cos(n\sigma) \frac{\partial}{\partial X_n^\mu}$$

$$= \frac{2}{\pi} \sum_{n=0}^{\infty} (e^{in\sigma} + e^{-in\sigma}) \frac{\partial}{\partial X_n^\mu}$$

Note, $X_{-n}^\mu = X_n^\mu$, thus

$$\frac{2}{\pi} \left(\sum_{n=0}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^\mu} + \sum_{n=-\infty}^0 e^{in\sigma} \frac{\partial}{\partial X_{-n}^\mu} \right)$$

$$\therefore \boxed{\frac{\delta}{\delta X^\mu(\sigma)} = \frac{2}{\pi} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^\mu}}$$

Subbing this into $\rho^\mu = -i\sqrt{2\alpha'} \pi \frac{\delta}{\delta X^\mu(\sigma)} + \frac{\dot{X}^\mu}{\sqrt{2\alpha'}}$

$$-i\sqrt{2\alpha'} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^\mu} + \dot{X}^\mu(\sigma)/\sqrt{2\alpha'}$$

For $\dot{X}^\mu$ we'd like to express this in terms of modes X_n^μ :

$$X^\mu(\tau, \sigma) = \sum_{n=-\infty}^{\infty} X_n^\mu(\tau) \cos(n\sigma)$$

$$= X_0^\mu(\tau) + \frac{2}{\pi} \sum_{n=1}^{\infty} X_n^\mu(\tau) e^{in\sigma}$$

$$\therefore \dot{X}^\mu(\tau, \sigma) = \frac{i}{2} \sum_{n \neq 0} X_n^\mu(\tau) n e^{in\sigma}$$

Note, we can include $n=0$ since

$$n=0 \text{ really is } = 0$$

$$\dot{X}^\mu(\tau, \sigma) = \frac{i}{2} \sum_{n=-\infty}^{\infty} X_n^\mu(\tau) e^{in\sigma}$$

$$\begin{aligned} \rho^\mu(\sigma) &= -i\sqrt{2\alpha'} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\partial}{\partial X_n^\mu} + \frac{i}{2} \sum_{n=-\infty}^{\infty} X_n^\mu n e^{in\sigma} \\ &= \sum_n \alpha_n^\mu e^{in\sigma} \end{aligned}$$

$$n \rightarrow -n$$

$$\boxed{\alpha_n^\mu = -i \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{-n}^\mu} + \frac{X_n^\mu n}{2\sqrt{2\alpha'}} \right)}$$

$$\frac{1}{\sqrt{2\alpha'}} \sum_{n=-\infty}^{\infty} \alpha_n^\mu e^{in\sigma}$$

* Virasoro ops: $L_n = \frac{1}{2} \sum_{m=-\infty}^{\infty} \alpha_m^\mu \alpha_{n-m}^\mu$

Normal ordering:

$$:\alpha_n^\mu \alpha_m^\nu: \equiv \begin{cases} \alpha_n^\mu \alpha_m^\nu, & \text{if } m > n \\ \alpha_m^\nu \alpha_n^\mu, & \text{if } n > m \end{cases}$$

i.e. we place the "larger" operator to the right

• Take a vacuum state $|0, p\rangle$:

$$\alpha_n^\mu |0, p\rangle = 0, \quad n \geq 1$$

$$\alpha_0^\mu |0, p\rangle = \sqrt{2\alpha'} p^\mu |0, p\rangle$$

Thus, for the present purpose only α_0^μ needs to be normal ordered. Because of the normal ordering issue the Witt Alg. becomes the full:

$$[L_n, L_m] = (n-m) L_{n+m} + \frac{D}{12} n(n^2-1) \delta_{n+m,0}$$

Take $m \rightarrow -n, \quad n \geq 0$

$$[L_n, L_{-n}] = 2n L_0 + \frac{D}{12} n(n^2-1) \mathbb{1}$$

$$\langle 0,0 | [L_n, L_{-n}] | 0,0 \rangle = \langle 0,0 | L_0 | 0,0 \rangle + \frac{D}{12} n(n^2-1) \langle 0,0 | 0,0 \rangle$$

$$\langle 0 | [L_n, L_{-n}] | 0 \rangle = \langle 0,0 | L_n L_{-n} | 0,0 \rangle$$

$$L_n = \frac{1}{2} : \sum_m \alpha_m^\mu \alpha_{n-m}^\mu :$$

$$\langle 0,0 | \frac{1}{2} : \sum_m \alpha_m^\mu \alpha_{n-m}^\mu : \frac{1}{2} : \sum_l \alpha_l^\sigma \alpha_{-n-l}^\sigma : | 0,0 \rangle$$

All states aside from the zero modes remain:

$$\frac{1}{4} \sum_{m=1}^{n-1} m(n-m) \langle 0,0 | 0,0 \rangle$$

$$L_0 = \frac{1}{2} \sum_m \alpha_m^\mu \alpha_{-m}^\mu = \frac{1}{2} \left(\sum_{m=1}^{\infty} \alpha_m^\mu \alpha_{-m}^\mu + \sum_{m=1}^{\infty} \alpha_{-m}^\mu \alpha_m^\mu \right)$$

$$= \frac{1}{2} \alpha_0^\mu \alpha_0^\mu \eta_{\mu\nu} + \frac{1}{2} \sum_{n=1}^{\infty} \alpha_{-n}^\mu \alpha_n^\mu + \sum_{n=1}^{\infty} \alpha_n^\mu \alpha_{-n}^\mu$$

$$= \frac{1}{2} \alpha_0 \cdot \alpha_0 + \sum_{n=1}^{\infty} (\alpha_{-n} \cdot \alpha_n)$$

* Imposing $L_n \psi = 0 \quad \forall n$ implies

$$\langle x^\mu(0, \epsilon) | L_n | \psi \rangle = \langle x^\mu(0, 0) | L_n | 0, 0 \rangle \langle 0, 0 | L_n | 0, 0 \rangle$$

$$\langle 0, 0 | \psi \rangle$$

$$\Rightarrow L_0 \psi = 0$$

$$(2\alpha L_0 + \frac{2}{12} D \alpha (n^2 - 1)) \psi = 0$$

$$n=0, \text{ (trivial)}$$

$$n=1: L_0 \psi = 0 \text{ (trivial)}$$

$$n=-1: L_0 \psi = 0 \text{ (trivial)}$$

$$n \geq 1: 2L_0 \psi = -\frac{D}{12} (n^2 - 1) \psi$$

$$\text{or } L_0 \psi \neq \frac{D}{24} (n^2 - 1) \psi = 0, \quad n \neq \pm 1$$

* Recall,

$$(\alpha_n^\mu)^\dagger = (\alpha_{-n}^\mu)$$

thus,

$$L_0 = \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + \sum_{n=1}^{\infty} \alpha_n^\mu \alpha_{-n\mu} \equiv \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + N$$

$$N \equiv \sum_{n=1}^{\infty} (\alpha_n^\mu)^\dagger \alpha_{n\mu}$$

$$L_1 = L_{-1} = \alpha_0^\mu \alpha_{1\mu} + \sum_{n=1}^{\infty} (\alpha_n^\mu)^\dagger \alpha_{n+1,\mu}$$

$$\alpha_0^\mu = -i\sqrt{2\alpha'} \frac{\partial}{\partial x_0^\mu} \equiv \sqrt{2\alpha'} p^\mu$$

$\Rightarrow \alpha_0^\mu \propto$ the momentum associated w/ the coordinates of M^D .

* Physical state conditions:

$$\psi[x^\mu(0)] \equiv \langle x^\mu(0) | \psi \rangle$$

which we express in terms of occupation # basis via Creation ops

$$\Psi[x^\nu(0)] = \{ \Psi(x^\nu) + iA_n^1(x^\nu) \alpha_1^{\nu\dagger} + iA_n^2(x^\nu) \alpha_2^{\nu\dagger} + \dots + \psi_{\text{vac}}(x^\nu) \alpha_1^{\nu\dagger} \alpha_1^{\nu\dagger} + \dots \} \Psi_0$$

• Treating zero modes & non-zero modes α differently. Ψ_0 refers to the non-zero modes and satisfy

$$\alpha_n^\mu \Psi_0 = 0, \quad n \geq 1$$

Ψ_0 : Vacuum state $\Psi_0 \equiv \langle 0, 0 | \Psi \rangle$

$$\therefore \alpha_n^\mu \Psi_0 \equiv \langle 0, 0 | \alpha_n^\mu | \Psi \rangle = 0$$

Since $\alpha_n^\mu |0, p\rangle = 0 \Leftrightarrow \langle 0, p | (\alpha_n^\mu)^\dagger = 0$
but $(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$ so $n \rightarrow -n$ shows this.

* Claim: The vacuum eq for $|0, p\rangle$ is a position "basis" is

$$\langle x^\mu(0) | 0, p \rangle = c \prod_{\nu=1}^{\infty} \exp\left(-\frac{n}{4\alpha'} x_n^\mu x_{n\mu}\right)$$

* Recall that for the harmonic oscillator

$$-\frac{1}{2m} \frac{d^2}{dx^2} \psi + \frac{1}{2} m \omega^2 x^2 \psi = E \psi$$

$$\xi \equiv \sqrt{m\omega} x$$

$$\psi(\xi) = \left[\frac{m\omega}{2^n n! \sqrt{\pi}} \right]^{1/2} H_n(\xi) e^{-\xi^2/2}$$

→ In this situation we have the Ground state

$\Psi_0(x) \equiv \langle x | 0 \rangle$, which we find via

$$\langle x | \hat{a} | 0 \rangle = \langle x | \left(\sqrt{\frac{m\omega}{2}} \tilde{x} + i \sqrt{\frac{1}{2m\omega}} \hat{p} \right) | 0 \rangle$$

but $\hat{a} | 0 \rangle = 0$

$$\langle x | 0 \rangle \times \sqrt{\frac{m\omega}{2}} + i \sqrt{\frac{1}{2m\omega}} \left(-i \frac{d}{dx} \langle x | 0 \rangle \right) = 0$$

$$\Psi_0 \times m\omega + \frac{d}{dx} \Psi_0 = 0$$

$$\Psi_0 = c \exp\left(-x^2 m\omega / 2\right)$$

(Classical
Harmonic Oscillator)

* For the open string : $\langle X^\mu(\sigma) | 0, p \rangle = \psi_0$

$$\langle X^\mu | \alpha_n^\mu | 0, p \rangle = 0$$

$$\langle X^\mu(\sigma) | (-i) \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{\mu n}} + \frac{n}{\sqrt{2\alpha'}} \frac{\hat{X}_n^\mu}{2} \right) | 0, p \rangle = 0$$

$$2\alpha' \frac{d}{dX_{\mu n}} \psi_0 + \frac{n}{2} X_n^\mu \psi_0 = 0$$

$$\therefore \left(n X_n^\mu X_{\mu n} / 4\alpha' \right)$$

$$\psi_{0,n} = c e$$

This is for a single n , to obtain all the full state

$$\psi_0 = \prod_{n=1}^{\infty} \psi_{0,n} = c \prod_{n=1}^{\infty} \exp \left[\frac{n}{4\alpha'} X_n^\mu X_{\mu n} \right]$$

(Vacuum state)

Physical states: $L_n \psi = 0$ $\left(\alpha_n^\mu = -i \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{\mu n}} + \frac{n}{\sqrt{2\alpha'}} \frac{X_n^\mu}{2} \right) \right)$

$$\begin{aligned} L_0 &= \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + \sum_{n=1}^{\infty} \alpha_n^\mu \alpha_{\mu n} \\ &= \frac{1}{2} \left(-i \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{0\mu}} \right) \right) \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{0\mu}} \right) + \dots \\ &= -\alpha' \left(\partial / \partial x^0 \right)^2 - \sum_{n=1}^{\infty} \left(\sqrt{2\alpha'} \frac{\partial}{\partial X_{\mu n}} + \frac{n}{\sqrt{2\alpha'}} \frac{X_n^\mu}{2} \right) \left(\dots \right) \end{aligned}$$

$$L_0 \psi = L_0 e^{\phi + \dots} \psi_0$$

$$\begin{aligned} (\partial_1 + x_1)(\partial_2 + x_2)f &= (\partial_1 + x_1)(\partial_2 f + x_2 f) \\ &= \partial_1 \partial_2 f + \partial_{12} f + x_2 \partial_1 f + x_1 \partial_2 f + x_1 x_2 f \end{aligned}$$

In acting on ψ only the first two terms contribute (consider why.....)

$$L_0 e^{\phi + \dots} \psi_0 = -(\partial^2 + 1)\phi + \dots$$

Each variables term by term for us to find

$$(\alpha' \partial^2 + 1) \phi = 0 = (\alpha' \partial^2 - (S-1)) A_m^S = (\partial^2 - 1) h_{\mu\nu} = \dots$$

as well as

$$\partial^\mu A_m^1 = 0 = -\sqrt{2\alpha'} \partial^\mu h_{\mu\nu} + A_\mu^2 = 2\sqrt{2\alpha'} \partial^\mu A_\mu^2 + h_\mu^4 =$$

where $\partial^2 \equiv (\partial/\partial x^\alpha)^2$

String Spectrum:

ϕ : Tachyon (negative mass) (Spin 0)
 $\rightarrow$ E.M should be $(\partial^2 - m^2)$ for "normal" mass

A_μ^1 : Spin - 1 - Vector

$h_{\mu\nu}$: Spin 2 Graviton

...

States at level n have a mass

$$m_n^2 = \left(\frac{2}{\alpha'}\right)(n-1)$$

• Particles of highest spin are centered in the distribution $\alpha_{-1}^{\mu_1} \dots \alpha_{-n}^{\mu_n} \epsilon_{\mu_1 \dots \mu_n}$ in ψ which have spin $J=n$

* We could impose $L_n \psi = 0$, $n \geq -1$, $\psi = e^{\phi_{-1}} \psi_0$
 obeys $\alpha' \partial^2 \phi = \dots$ i.e. where the "1s" are removed

$$L_{-1} \psi = 0 \Rightarrow \partial_\mu \psi = 0 = \partial_\mu A_\nu + \partial_\nu A_\mu$$

Soln.: $\phi = (\text{constant})$

Remaining fields vanish

$$\begin{aligned} \psi[x^\mu(\sigma)] &= e^\phi \psi_0 = e^\phi \langle x^\mu(\sigma) | 0, p \rangle \\ &= \langle x^\mu(\sigma) | p \rangle \end{aligned}$$

Some insight into the Masses found:

* Classically $T_{\alpha\beta} = 0 \Leftrightarrow L_m = 0, \quad m \in \mathbb{Z}$

$$L_0 = \tilde{L}_0 = 0.$$

$$\therefore (\text{Mass})^2 \equiv M^2 = -p_\mu p^\mu$$

$$L_0 = \sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n + \alpha' p^2 = 0$$

$$M^2 = \frac{1}{\alpha'} \left(\sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n \right) \quad (\text{open string})$$

Physical states: $(L_0 - a)|\phi\rangle = 0$

$\leadsto$ Mass operator

$$\hat{N} \equiv \sum_{n=1}^{\infty} \alpha_{-n} \cdot \alpha_n$$

$$\alpha' M^2 = \hat{N} - a$$

Since we require $a=1$,

$$\alpha' M^2 = \hat{N} - 1$$

Ground state: $\hat{N} = 0$

$$\alpha' \hat{M}^2 = -1 \quad \text{Tachyon } \phi$$

$$\hat{N} = 1: \quad \alpha' \hat{M}^2 = 0$$

$$\hat{N} = 2: \quad \alpha' \hat{M}^2 = 1$$

* $X^\mu(\sigma, \tau)$ contains an ∞ of possible neg.

|| states, each associated to α_n^μ

$$[\alpha_n^\mu, \alpha_n^\mu] = -n.$$

$$\text{ex: } \alpha_{-n}^\mu |0, p\rangle = (\alpha_n^\mu)^\dagger |0, p\rangle$$

$$[\langle 0, p | \alpha_n^\mu \alpha_{-n}^\mu | 0, p \rangle = -n \langle 0, p | 0, p \rangle$$

- Physical states must have a positive norm if we are to interpret the formalism as a probability

(Prove later)

Thm: [No ghost Theorem] For space-time dim $D \leq 26$ states ψ which satisfy the eqs $L_n \psi = 0$, $n \geq 1$, $(L_0 - 1)\psi = 0$ have positive norm

* Bosonic string inconsistent in $D \neq 26$

- For $D \leq 26$, the norms of the physical states are not all ≥ 0

ex: $|s\rangle \equiv L_{-1}|\Omega\rangle$ / $L_0|\Omega\rangle = 0$
 $L_n|\Omega\rangle = 0, n \geq 1$

Claim: $L_0|s\rangle = |s\rangle$

Proof: $L_0 L_{-1}|\Omega\rangle = (L_{-1} + 0)|\Omega\rangle = |s\rangle \checkmark$

- Let $\langle\psi|$ be any other physical state, then we find

$$\langle\psi|s\rangle = \langle\psi|L_{-1}|\Omega\rangle = (\langle\Omega|L_1|\psi\rangle)^*$$

Since $L_1|\psi\rangle = 0$ (physical state requirement)

$$\langle s|s\rangle = 0$$

* Let $D = 26$:

$$[L_n, L_m] = (n-m)L_{n+m} + \frac{D}{12}n(n^2-1)\delta_{n+m,0}$$

$$\Rightarrow [L_n, L_m] = (n-m)L_{n+m} + \frac{13}{6}n(n^2-1)\delta_{n+m,0}$$

- Take $|s'\rangle \equiv (L_{-2} + 6L_{-1}^2)|\Omega'\rangle$

$$L_n|\Omega'\rangle, n \geq 1$$

$$(L_0 + 1)|\Omega'\rangle = 0$$

- Demanding $|s'\rangle$ be physical

$$L_1|s'\rangle = 0$$

$$L_1(L_{-2} + 6L_{-1}L_{-1})|\Omega'\rangle$$

$$\left. \begin{array}{l} n \geq 0 \\ L_n|\psi\rangle = 0 \\ \text{AND} \\ (L_0 - 1)|\psi\rangle = 0 \end{array} \right\}$$

$$\begin{aligned}
 &= (3L_{-1} + b2L_0L_{-1})|\Omega'\rangle \\
 &= (3L_{-1} + 2bL_{-1})|\Omega'\rangle \\
 &\text{or } (3 - 2b)L_{-1}|\Omega'\rangle \\
 &\Rightarrow b = +\frac{3}{2}
 \end{aligned}$$

Mass-shell condition

* Zero-norm physical states $n \geq 1$
 $L_{-n_1} \dots L_{-n_p} |\Omega''\rangle, \quad L_n |\Omega''\rangle = 0$
 $L_0 |\Omega''\rangle = \sum_{q=1}^p (n_q + 1) |\Omega''\rangle$

SPECTRUM: Vector particle A^μ subject to Maxwell's Eq's:

$$\partial_\mu F^{\mu\nu} = \partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$$

$$\partial^2 A^\nu - \partial_\mu \partial^\mu A^\nu = 0$$

• Choose the Gauge $\partial_\mu A^\mu = 0$
 $\partial^2 A^\nu = 0$

$$A^\mu(x^\sigma) = \int \frac{d^4k}{(2\pi)^4} (a^\mu_\lambda e^{-ikx} + a^{\mu\dagger}_\lambda e^{ikx})$$

$$a^\lambda_\lambda \equiv \sum_{\lambda=0}^3 a^\lambda e^\lambda$$

λ : Polarization index:

$$A^\mu(x)|0\rangle \sim \epsilon^\mu_\lambda a^\dagger_\lambda |0\rangle$$

which we find from the string spectrum if we identify

$$a^\mu_{-1} e^{ik_\mu x^\sigma} \equiv a^{\mu\dagger}(k)$$

$$a^\mu_1 e^{-ik_\mu x^\sigma} \equiv a^\mu(k)$$

$$A^\mu_m(x) = \epsilon^\mu_m e^{-ik_\mu x^\sigma}$$

• $\exp(-ik_\mu x^\sigma)$ is required to meet the momentum

K^μ into the vacuum $|0\rangle$:

$$\exp(ik^\mu q_\mu)|0\rangle = |K\rangle$$

The E.O.M:

$$\begin{aligned}\partial^2 A^\mu &= \partial^2 \int \frac{d^3K}{(2\pi)^3 2\omega_K} (a^\lambda G_\lambda^\mu e^{-iKx} + a^{\dagger\lambda} \epsilon_\lambda^{\mu\nu} e^{iKx}) \\ &= 0 \\ &= \int \frac{d^3K}{(2\pi)^3 2\omega_K} -K^2 (a^\lambda \epsilon_\lambda^\mu e^{-iKx} + \dots)\end{aligned}$$

and so we conclude

$$K^2 = 0 \quad (\partial^2 A^\mu = 0)$$

* The Gauge charge tells us (trivially)

$$K^\mu \epsilon_\mu = 0 \quad (\partial^\mu A_\mu = 0)$$

* In the Lorentz frame (Massless particle)

$$K^\mu = (K, 0, \dots, K)$$

$$K^\mu G_\mu = K G_0 + K G_D = 0$$

$$\text{If } K \neq 0, \text{ then } G_0 = -G_{D-1}$$

$$\epsilon_\mu (\alpha^\mu)_i^\dagger |0, K\rangle = \begin{cases} (\alpha_i^\dagger)^\dagger |0, K\rangle, & i=1, \dots, D-2 \\ (-\alpha_i^\dagger)^\dagger + (\alpha_{i^{D-1}}^\dagger)^\dagger |0, K\rangle \\ (\alpha_i^\dagger)^\dagger + (\alpha_{i^{D-1}}^\dagger)^\dagger |0, K\rangle \end{cases}$$

* Can remove zero norm state via the residual gauge symmetry

$$G_\mu \rightarrow G_\mu + K_\mu \lambda$$

- Leaves us with $(D-2)$ physical pos. def. norm states: matching our vector rep of $SO(D-2)$ (Little group)

* For $D=26$: the # of physical states w $\alpha \cdot \alpha > 0$ at level n , which have

$$\prod_{n=1}^{\infty} d\alpha_n x^n = \prod_{n=1}^{\infty} \frac{1}{(1-x^{2n})^{24}} = 1 + 24x + \dots$$

$\underbrace{\quad}_{\phi} \quad \underbrace{\quad}_{A_{\mu}}$

ϕ : Has 4 D.O.F

A_{μ} : Has 24 D.O.F (in $D=26$)

$$\begin{aligned}
 P_M &= \frac{\dot{x}^M}{\sqrt{2\alpha'}} = -i \frac{\partial}{\partial x^M} \\
 J^{MN} &= \frac{1}{2\pi\alpha'} \int_0^{2\pi} d\sigma (x^M \partial_\tau x^N - x^N \partial_\tau x^M) \\
 &= \frac{1}{2\pi\alpha'} \sqrt{\frac{\alpha'}{2}} \int_{-\pi}^{\pi} d\sigma (x^M p^N - x^N p^M) \\
 &= \frac{1}{\sqrt{2\alpha'}} (x_0^M \alpha_0^N - x_0^N \alpha_0^M) - i \sum_{n=1}^{\infty} (\alpha_{-n}^M \alpha_n^N - \alpha_{-n}^N \alpha_n^M) / n
 \end{aligned}$$

The Closed String:

We have $X^M, P^M, \alpha_n^M, \bar{\alpha}_n^M$ with C.O.M coords (q^M, p^M) . Recall

$$\alpha_0^M = \bar{\alpha}_0^M = \sqrt{\frac{\alpha'}{2}} p^M$$

$$\{\alpha_n^M, \alpha_m^N\} = -i n \delta_{n+m,0} \eta^{MN}, \quad \{\alpha_n^M, \bar{\alpha}_m^N\} = 0$$

$$\{\bar{\alpha}_n^M, \bar{\alpha}_m^N\} = -i n \delta_{n+m,0} \eta^{MN}$$

→ Classical: From here we use Dirac Quant.

$$\begin{aligned}
 [\alpha_n^M, \alpha_m^N] &= n \delta_{n+m,0} \eta^{MN} & [\alpha_n^M, \bar{\alpha}_m^N] &= 0 \\
 [\bar{\alpha}_n^M, \bar{\alpha}_m^N] &= n \delta_{n+m,0} \eta^{MN} & [q^M, p^N] &= i \eta^{MN}
 \end{aligned}$$

For $n=0$:

$$\begin{aligned}
 \int_0^{2\pi} d\sigma \{P^M(\sigma), P^N(\sigma)\} &= \int_0^{2\pi} d\sigma \pi \eta^{MN} \delta(\sigma-\sigma') d\sigma' \\
 \int_0^{2\pi} d\sigma \{P^M(\sigma), \sqrt{2\alpha'} \pi P^N\} &= 2\pi \eta^{MN} \\
 \int_{-\pi}^{\pi} d\sigma \{P^M(\sigma), \sqrt{2\alpha'} \pi P^N\} &= (2\pi)^2 \eta^{MN} \\
 \int_{-\pi}^{\pi} d\sigma \left\{ \frac{1}{2\alpha'} \left(\int d\sigma \dot{x}^2 + x^M \right), \sqrt{2\alpha'} \pi P^N \right\} &= (2\pi)^2 \eta^{MN} \\
 \left\{ \frac{1}{2\alpha'} (q^M + \sqrt{2\alpha'}^2 \alpha_0^M), \sqrt{2\alpha'} \pi P^N \right\} &
 \end{aligned}$$

$$\Rightarrow \{q^\mu, p^\nu\} = \eta^{\mu\nu}$$

$$\circ \circ [q^\mu, p^\nu] = i\eta^{\mu\nu}$$

* Reality conditions of $(\hat{P}^\mu)^\dagger = \hat{P}^\mu$ implies:

$$(\alpha_n^\mu)^\dagger = \alpha_{-n}^\mu$$

$$\bar{\alpha}_{-n}^\mu = (\bar{\alpha}_n^\mu)^\dagger$$

$$q^\mu \dagger = q^\mu \text{ and } p^\mu \dagger = p^\mu$$

$$\# \hat{P}^\mu = \frac{1}{2\pi\alpha'} \hat{X}^\mu, \quad \hat{P}^\mu = -i \frac{\delta}{\delta X_\mu(\sigma)}$$

$$\hat{P}^\mu = \frac{1}{\sqrt{2\alpha'}} (2\pi\alpha' \hat{P}^\mu|_0 + \dot{X}^\mu|_0) = \sum_{n=-\infty}^{\infty} \alpha_n^\mu e^{-in\sigma}$$

$$= (\sqrt{2\alpha'} \pi \hat{P}^\mu + \frac{\dot{X}^\mu}{\sqrt{2\alpha'}})$$

$$\hat{P}^\mu = (\sqrt{2\alpha'} \pi \hat{P}^\mu - \dot{X}^\mu/\sqrt{2\alpha'}) = \sum_{n=-\infty}^{\infty} \bar{\alpha}_n^\mu e^{in\sigma}$$

$$(3.1.6) \text{ \& } (2.1.37)$$

$$\dot{X}^\mu(\tau, \sigma) = \sum_{n=-\infty}^{\infty} e^{in\sigma} \dot{X}_n^\mu(\tau)$$

$$\frac{\delta}{\delta X_\mu(\sigma)} = \sum_{n=-\infty}^{\infty} \frac{\delta X_n^\mu(\sigma)}{\delta X_\mu(\sigma)} \frac{\delta}{\delta X_n^\mu(\sigma)} = \frac{1}{\pi} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\delta}{\delta X_n^\mu}$$

$$\sqrt{2\alpha'} \pi \left(\frac{1}{\pi} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\delta}{\delta X_n^\mu} \right) + \sum_{n=-\infty}^{\infty} -in e^{in\sigma} \dot{X}_n^\mu(\tau)/\sqrt{2\alpha'}$$

$$= \sqrt{2\alpha'} \frac{1}{2} \sum_{n=-\infty}^{\infty} e^{in\sigma} \frac{\delta}{\delta X_n^\mu}$$

$$\sum_{n=-\infty}^{\infty} e^{in\sigma} \left[\sqrt{\frac{\alpha'}{2}} \frac{\delta}{\delta X_n^\mu} + \frac{in \dot{X}_n^\mu}{\sqrt{2\alpha'}} \right] = \sum_{n=-\infty}^{\infty} \alpha_n^\mu e^{in\sigma}$$

$$\alpha_n^\mu = -i\sqrt{\frac{\alpha'}{2}} \frac{\delta}{\delta X_n^\mu} - \frac{in \dot{X}_n^\mu}{\sqrt{2\alpha'}}$$

$$\bar{\alpha}_n^\mu = -i\sqrt{\frac{\alpha'}{2}} \frac{\delta}{\delta X_{-n}^\mu} - \frac{in \dot{X}_{-n}^\mu}{\sqrt{2\alpha'}}$$

Plugging these into $[\alpha_n^\mu, \alpha_m^\nu]$ we obtain the above.

* The Virasoro ops:

$$L_n = \frac{1}{2} \circ \sum_{m=-\infty}^{\infty} \alpha_m^\mu \alpha_{n-m}^\mu \circ \eta_{\mu\nu}$$

$$\bar{L}_n = \frac{1}{2} \circ \sum_{m=-\infty}^{\infty} \bar{\alpha}_m^\mu \bar{\alpha}_{n-m}^\mu \circ \eta_{\mu\nu}$$

Physical state Conditions:

$$L_n \psi = \bar{L}_n \psi = 0, \quad n \geq 1$$

$$(L_0 + \bar{L}_0 - 2) \psi = 0$$

$$(L_0 - \bar{L}_0) \psi = 0$$

$\rightarrow$ Interpretation of $(L_0 - \bar{L}_0) \psi = 0$

$$L_0 = \frac{1}{2} : \sum_{n=-\infty}^{\infty} \alpha_n^\mu \alpha_{-n}^\mu : = \frac{1}{2} \alpha_0^\mu \alpha_{0\mu} + \sum_{n=1}^{\infty} \alpha_{-n}^\mu \alpha_n^\mu \eta_{\mu\nu}$$

$$(2.2.3) \text{ \& } (2.1.37)$$

$$x^\mu = \sum e^{in\sigma} \alpha_n^\mu(\tau)$$

$$(L_0 - \bar{L}_0) = \frac{2}{4\pi} \int_{-\pi}^{\pi} d\sigma ((\dot{x}^2) - (\dot{\bar{x}}^2)) = \frac{1}{4\pi} \int_{-\pi}^{\pi} d\sigma 2 \dot{x}^\mu(\sigma) \frac{\delta}{\delta \dot{x}^\mu(\sigma)}$$

$2\pi\epsilon(L_0 - \bar{L}_0)$ implements the charge $\sigma \rightarrow \sigma + 2$ and so $(L_0 - \bar{L}_0) \psi = 0$ tells us there is no preferred pt. along the length of the closed string.

$$\psi[x^\mu(\sigma)] \equiv \langle x^\mu(\sigma) | \psi \rangle$$

$$\alpha_n^\mu \psi_0 = \bar{\alpha}_n^\mu \psi_0 = 0, \quad n \geq 1$$

$$(\alpha' \partial^2 + 4) \psi = 0$$

$$\partial^2 h_{\mu\nu} = 0$$

$$\partial^\mu h_{\mu\nu} = 0 = \partial^\nu h_{\mu\nu}$$

(c) Unoriented Strings: $\sigma \rightarrow -\sigma$

is a symmetry of the theory

is a symmetry of the theory

- Every moving wave around the closed string
which depends on $\tau + \sigma$ we have the same
amplitude which depends on $\tau - \sigma$.
- Equal amplitudes moving in opposite directions.
- This imposes

$$\alpha_n^M \longleftrightarrow \bar{\alpha}_n^M$$

(ii) Oriented String: Amplitudes for X_L & X_R are
unrelated.

BRST Symmetry